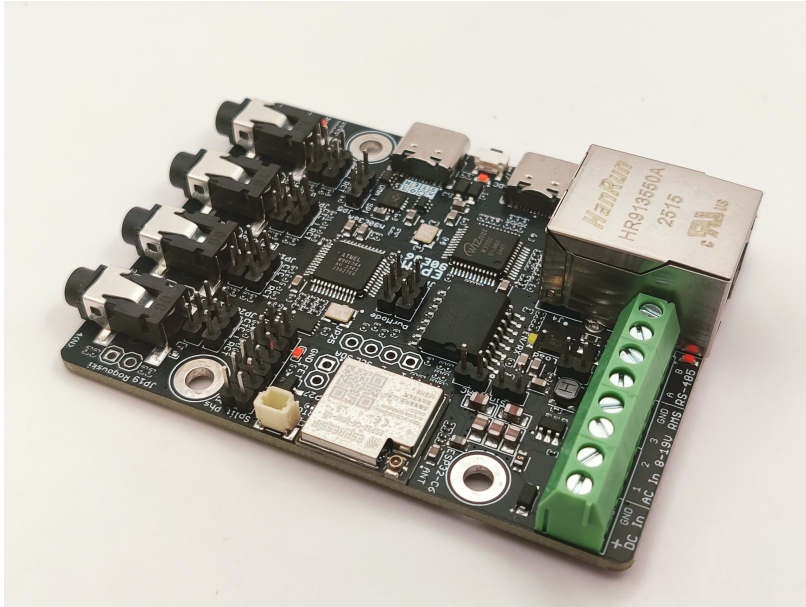


DTM-EMS

Precision Distribution-Transformer Monitoring with Neutral, Phase & Leakage Insight



DTM-EMS pairs the proven **Ditronix EPEM / IPEM** precision-metering platform (Microchip ATM90E36A, $\pm 0.1\%$ polyphase) with an open-source insights layer — delivering transformer-grade monitoring of neutral, phase and leakage, plus dynamic energy-policy enforcement at the secondary distribution edge.

What Sets DTM-EMS Apart:

- ▶ **Polyphase Precision Metering** : Ditronix ATM90E36A, $\pm 0.1\%$ active-energy accuracy, 3-phase 4-wire with a dedicated neutral channel.
- ▶ **Neutral / Phase / Leakage Insight** : measures neutral current and phase imbalance directly, and supports non-intrusive load monitoring (NILM).
- ▶ **Open Edge Integration** : the IPEM PiHat exposes the meter over SPI / 40-pin header to the NESL co-processor for dynamic policy enforcement (stackable up to 64 boards).

DTM-EMS gives operators a precise, intelligent view of the secondary transformer — neutral, phases and leakage — and the means to enforce energy policy at the edge.

Intelligent

- ▶ Neutral / phase / leakage + NILM insights
- ▶ ATM90E36A polyphase, $\pm 0.1\%$ active energy
- ▶ Dynamic edge policy enforcement

Interoperable

- ▶ EPEM Ethernet (W5500) + RS-485 Modbus
- ▶ IPEM PiHat: SPI / 40-pin header to any MCU / Pi
- ▶ WiFi 6 / Thread / Zigbee (EPEM)

Reliable

- ▶ Proven Ditronix metering hardware
- ▶ Onboard RTC, temperature & EEPROM
- ▶ Stackable up to 64 boards

PRODUCT	
Product name	DTM-EMS (10Power X-EMS, supplied by NESL)
Metering platform	Ditronix EPEM / IPEM PiHat (Microchip ATM90E36A)
Firmware	meshEMS-integrated open-source insights & policy layer
METROLOGY	
Metering IC	Microchip ATM90E36A polyphase (ATM90E32 option)
Channels	3 voltage inputs up to 4 CT: 3 phase + 1 neutral (clamp or Rogowski)
Accuracy	$\pm 0.1\%$ active energy $\pm 0.2\%$ reactive energy
INSIGHTS	
Insights engine	Neutral phase imbalance leakage NILM (non-intrusive load monitoring)
Measured	RMS V / I P / Q / S harmonics PF phase angle frequency bidirectional energy
POLICY ENFORCEMENT	
Edge enforcement	Dynamic energy-policy enforcement at the transformer edge
Coordination	Site-networked EMS + DER coordination
INTEGRATION & NETWORKING	
Co-processor link	IPEM PiHat: SPI + 40-pin Pi header (any MCU / Pi, stackable $\times 64$) EPEM SPI is internal (ESP32-C6)
Networking	Ethernet (WIZnet W5500, RJ45) RS-485 Modbus WiFi 6 / Thread / Zigbee (EPEM)
Onboard	RTC (DS3231) temperature (TMP102) EEPROM RGB status
POWER & MECHANICAL	
Power supply	EPEM: 5 V USB-C or 4.5–20 V DC IPEM: 8–12 V AC supply (SELV)
Dimensions	EPEM 85 $\times$ 55 mm IPEM Raspberry Pi HAT form factor
STANDARDS & POSITIONING	
Grid position	Secondary distribution transformer (DTM)
Capabilities	Polyphase precision metering neutral / phase / leakage insight edge policy enforcement

DTM-EMS combines the Ditronix EPEM / IPEM platform (ATM90E36A) with an open-source insights and policy layer by New Energy Solutions Lab for 10Power. Metering specifications per Ditronix (DitroniX.net). Customizable with added modules and open or licensed code stacks; items marked **[to confirm]** to be confirmed.